Quantum Physics, 2025 Fall, BME
Lecture 06 - Composite quantum systems

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Contents

1 Composite quantum systems are described by vectors in the tensor product space **1**
- 1.1 Tensor product of two complex vectors . . . 2
- 1.2 Entanglement . . . 2
- 1.3 Properties of the tensor product of vectors . . . 3
- 1.4 Tensor products of operators . . . 4

2 Time evolution of composite systems **5**

3 Local measurement in a composite system **6**

4 Further examples for composite systems **6**
- 4.1 Qubit and oscillator . . . 6
- 4.2 Two coupled harmonic oscillators . . . 7

1 Composite quantum systems are described by vectors in the tensor product space

1. In classical mechanics, sometimes we talk about 'composite systems', consisting of at least two particles that interact with each other. E.g. Sun and Earth interacting by gravity, or two point masses interacting via a connecting spring, etc. Such an interaction leads to coupled equations of motions for the coordinates and velocities of the two (or more) interacting objects.

2. In quantum physics, the state of an isolated quantum system is described by a complex state vector. E.g., the spin of a trapped electron, or a qubit, is described by $|\psi\rangle \in \mathbb{C}^2$.

3. How to describe two qubits, e.g., the spins of two electrons, each trapped in a different trap? How to describe two interacting quantum harmonic oscillators?

1.1 Tensor product of two complex vectors

1. Given two vector spaces $V = \mathbb{C}^n$ and $W = \mathbb{C}^m$, we can build a greater vector space $V \otimes W$ from those. The new vector space $V \otimes W = \mathbb{C}^n \otimes \mathbb{C}^m$ is called the *tensor product* of the two original vector spaces, it has dimension $n \cdot m$, and it can be thought of as $\mathbb{C}^{n \cdot m}$.

2. We can embed a pair of vectors $(|v\rangle, |w\rangle)$, where $|v\rangle \in V = \mathbb{C}^n$ and $|w\rangle \in W = \mathbb{C}^m$, into $V \otimes W = \mathbb{C}^{n \cdot m}$, as follows:

$$(|v\rangle, |w\rangle) \mapsto \begin{bmatrix} v_1 w \\ v_2 w \\ \dots \\ v_n w \end{bmatrix} = \left[\begin{array}{c} v_1 w_1 \\ \dots \\ v_1 w_m \\ \hline v_2 w_1 \\ \dots \\ v_2 w_m \\ \hline \dots \\ \hline v_n w_1 \\ \dots \\ v_n w_m \end{array}\right] = |v\rangle \otimes |w\rangle . \tag{1}$$

3. The resulting vector in $\mathbb{C}^{n \cdot m}$ is called the tensor product of $|v\rangle$ and $|w\rangle$, and is denoted in a few different ways:

$$|v\rangle \otimes |w\rangle \equiv |v\rangle |w\rangle \equiv |v, w\rangle \equiv |vw\rangle . \tag{2}$$

4. Special case of Eq. (1) for two qubits ($V = W = \mathbb{C}^2$): the tensor product of $|v\rangle \in \mathbb{C}^2$ and $|w\rangle \in \mathbb{C}^2$ is embedded into $\mathbb{C}^4$ as follows:

$$|v\rangle \otimes |w\rangle \equiv \begin{bmatrix} v_1 w \\ v_2 w \end{bmatrix} \equiv \begin{bmatrix} v_1 w_1 \\ v_1 w_2 \\ v_2 w_1 \\ v_2 w_2 \end{bmatrix} . \tag{3}$$

5. Exercise: For two qubits, with $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$, compute the four-component complex vectors representing these tensor product vectors:
   (a) $|0\rangle \otimes |0\rangle$,
   (b) $|0\rangle \otimes |1\rangle$,
   (c) $|0\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$,
   (d) $\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |0\rangle$.

   Compare the results (c) and (d). Is the tensor product 'commutative'?

1.2 Entanglement

1. The set $\mathbb{C}^{n \cdot m}$ is a vector space. However, there are vectors in $\mathbb{C}^{n \cdot m}$ that cannot be obtained as a tensor product of a vector in $\mathbb{C}^n$ and a vector in $\mathbb{C}^m$.

2. Proof, by example: Take $(1,0,0,1)^T \in \mathbb{C}^4$. Can it be written as $|v\rangle \otimes |w\rangle$ with $|v\rangle, |w\rangle \in \mathbb{C}^2$? To see this, try to solve the corresponding equation:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} v_1 w_1 \\ v_1 w_2 \\ v_2 w_1 \\ v_2 w_2 \end{bmatrix} \tag{4}$$

The second equation implies that $v_1 = 0$ or $w_2 = 0$. If $v_1 = 0$, then the right hand side of the first equation is zero. The left hand side is 1, hence we have a contradiction. If $w_2 = 0$, then the right hand side of the fourth equation is zero. The left hand side is 1, hence we have a contradiction again. QED.

3. Some four-component complex vectors can be written as a single tensor product of two two-component complex vectors. Those are called *product states*. Examples: see exercise at the end of the previous section.

   Some four-component complex vectors cannot be written as a tensor product of two two-component complex vectors. Those are called *entangled* states. Examples:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \; |\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \; |\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \; |\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}. \tag{5}$$

   For a visualisation of the difference between product states and entangled states, see Figure 1.

4. Entanglement is an important characteristics of quantum physics, which distinguishes quantum physics from classical physics and probability theory.

5. Even though we illustrated entanglement with the example of two qubits, the notion of entanglement naturally generalizes to quantum systems described by higher-dimensional vector spaces, and for composite systems with more than two subsystems.

6. Although not all states in the tensor product space are product states, all states can be obtained as linear combinations of product states. Proof: the tensor products of the canonical basis vectors of $\mathbb{C}^n$ and $\mathbb{C}^m$ form a basis in $\mathbb{C}^{n \cdot m}$.

1.3 Properties of the tensor product of vectors

1. For a scalar z, and vectors $|v\rangle \in V$ and $|w\rangle \in W$, it holds that

$$z\,(|v\rangle \otimes |w\rangle) = (z\,|v\rangle) \otimes |w\rangle = |v\rangle \otimes (z\,|w\rangle)\,. \tag{6}$$

2. For vectors $|v_1\rangle, |v_2\rangle \in V$ and $|w\rangle \in W$, it holds that

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle\,. \tag{7}$$

3. Conversely, for vectors $|v\rangle \in V$ and $|w_1\rangle, |w_2\rangle \in W$, it holds that

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle\,. \tag{8}$$

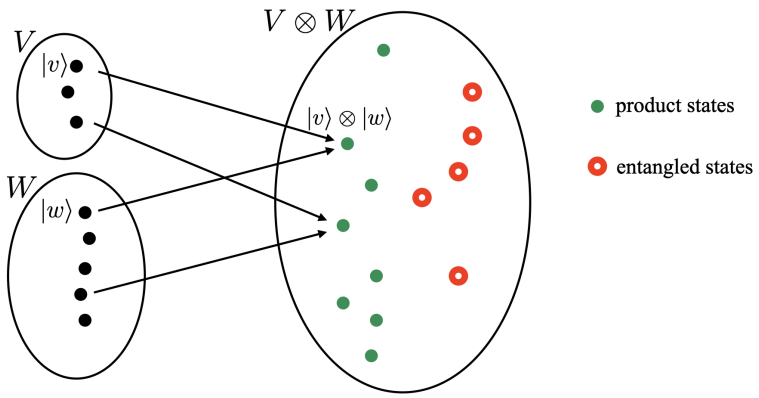


Figure 1: Creating the tensor product vector space $V \otimes W$ from two vector spaces V and W. Some vectors in $V \otimes W$ are created from pairs of vectors in V and W. These are *product states*, green points. Other vectors in $V \otimes W$ are linear combinations (or superpositions) of tensor-product vectors. Those are *entangled states*, red circles.

1.4 Tensor products of operators

1. We can also define the tensor product $A \otimes B$ of two operators A and B, where A acts on V and B acts on W. For example, if we talk about two qubits, then we can represent $A \otimes B$ as a 4×4 complex matrix:

$$A \otimes B \equiv \begin{bmatrix} A_{11}B & A_{12}B \\ A_{21}B & A_{22}B \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} & A_{11}B_{12} & A_{12}B_{11} & A_{12}B_{12} \\ A_{11}B_{21} & A_{11}B_{22} & A_{12}B_{21} & A_{12}B_{22} \\ A_{21}B_{11} & A_{21}B_{12} & A_{22}B_{11} & A_{22}B_{12} \\ A_{21}B_{21} & A_{21}B_{22} & A_{22}B_{21} & A_{22}B_{22} \end{bmatrix} \tag{9}$$

2. Exercise: Compute the 4×4 complex matrices representing these tensor products:
   (a) $I \otimes I$,
   (b) $I \otimes X$,
   (c) $X \otimes Z$,
   (d) $X \otimes Y$.

3. The above examples describe the composite system of two qubits. The same concepts are straightforwardly generalized to systems described by higher-dimensional Hilbert spaces, and to more than two subsystems.

4. Python Demo: show the command implementing Kronecker product for vectors and for matrices. How to implement the case of 3 or more subsystems?

5. *A property of the tensor product of linear operators.* If A is a linear operator acting on V, B is a linear operator acting on W, and $|v\rangle \in V$ and $|w\rangle \in W$, then it holds that:

$$(A \otimes B)\,(|v\rangle \otimes |w\rangle) = (A\,|v\rangle) \otimes (B\,|w\rangle)\,. \tag{10}$$

6. *Scalar product on the tensor product space.* The tensor product space $\mathbb{C}^{n\cdot m}$ has a canonical scalar product $(.,.)$. Its relation to the scalar product of the composing spaces is as follows:

$$\left(\sum_i a_i \left|v_i\right\rangle \otimes \left|w_i\right\rangle, \sum_j b_j \left|v'_j\right\rangle \otimes \left|w'_j\right\rangle\right) = \sum_{ij} a_i^* b_j \left\langle v_i|v'_j\right\rangle \left\langle w_i|w'_j\right\rangle. \tag{11}$$

7. Here I focused on tensor products related to $\mathbb{C}^n$ vector spaces. The tensor product is a more general concept, applicable to other vector spaces. A slightly more general and abstract 2-page description is given in Nielsen-Chuang chapter 2.1.7 "Tensor products".

2 Time evolution of composite systems

1. Example: two spins in a magnetic field, interacting via 'Ising interaction':

$$H = H_{\mathrm{L}} + H_{\mathrm{R}} + H_{\mathrm{int}}, \tag{12}$$

where

$$H_{\mathrm{L}} = \frac{1}{2} g\mu_B B \left(\sigma_z \otimes I\right), \tag{13a}$$

$$H_{\mathrm{R}} = \frac{1}{2} g\mu_B B \left(I \otimes \sigma_z\right), \tag{13b}$$

$$H_{\mathrm{int}} = J\sigma_z \otimes \sigma_z. \tag{13c}$$

Assume that the initial state is $\left|\psi_0\right\rangle = \frac{1}{\sqrt{2}}\left(\left|\downarrow\right\rangle + \left|\uparrow\right\rangle\right) \otimes \left|\downarrow\right\rangle$. Compute time evolution.

2. An alternative, more compact, often used notation for the Hamiltonian:

$$H_{\mathrm{L}} = \frac{1}{2} g\mu_B B\sigma_{z,L}, \tag{14a}$$

$$H_{\mathrm{R}} = \frac{1}{2} g\mu_B B\sigma_{z,R}, \tag{14b}$$

$$H_{\mathrm{int}} = J\sigma_{z,L}\sigma_{z,R}. \tag{14c}$$

3. *Eigenvalues and eigenvectors of a noninteracting composite system are expressed from those of the subsystems.*

Consider a two-component composite quantum system that is *noninteracting*. This means that the Hamiltonian of the composite system is $H = H_A + H_B$, where $H_A = A \otimes I$ and $H_B = I \otimes B$.

Assume that the eigenvalue-eigenvector pairs of A and B are known: $(\lambda_i, \left|v_i\right\rangle)$ for A ($i \in \{1, \ldots, n\}$), and $(\nu_j, \left|w_j\right\rangle)$ for B ($j \in \{1, \ldots, m\}$).

Then, the eigenvalues-eigenvalue pairs of H are

$$\left(\lambda_i + \nu_j, \left|v_i\right\rangle \otimes \left|w_j\right\rangle\right), \text{ where } i \in \{1, \ldots, n\}, j \in \{1, \ldots, m\}. \tag{15}$$

4. Exercise: express the four eigenvalue-eigenvector pairs of H of Eq. (12), assuming $H_{\rm int} = 0$.

5. Exercise: express the four eigenvalue-eigenvector pairs of H of Eq. (12), keeping $H_{\rm int}$ in the Hamiltonian.

3 Local measurement in a composite system

1. It often happens that the subsystems of a composite system have been interacting in the past, but now they are far apart, and we have access to only one of the subsystems.

2. In such a situation, we can do a *local measurement* on the subsystem we have access to. That is a measurement of a local observable having the form $M \otimes 1$.

3. Example: Two qubits in a general state, and we measure the first one in the computational basis.

4. It is straightforward to express the measurement statistics and post-measurement states from Postulate 3 (the measurement postulate, Lecture 02), although the formulas are not very instructive.

5. The spectral decomposition of $M \otimes I$ can be expressed from that of M. E.g., if M has a nondegenerate spectrum, then it has n different eigenvalues, and its spectral decomposition is $M = \sum_{j=1}^{n} \lambda_j P_j$. Correspondingly, $M \otimes I$ also has the same n different eigenvalues, each of them being m-fold degenerate. The spectral decomposition reads:

$$M \otimes I = \sum_{j=1}^{n} \lambda_j P_j \otimes I = \sum_{j=1}^{n} \lambda_j P_j^c, \tag{16}$$

where the superscript c signals that $P_j^c \equiv P_j \otimes I$ is a projector on the composite system.

6. The measurement statistics and the post-measurement state can be expressed as we have seen in Postulate 3:

$$p_j = p(\lambda_j) = \langle\psi|P_j^c|\psi\rangle\,, \tag{17}$$

$$|\psi_{\rm post}\rangle = \frac{P_j^c\,|\psi\rangle}{\sqrt{p_j}}. \tag{18}$$

4 Further examples for composite systems

4.1 Qubit and oscillator

1. Setup: Magnet pointing along z, positioned at $x = 0$, spin located along x axis at position x_0, as shown in Fig. 2.

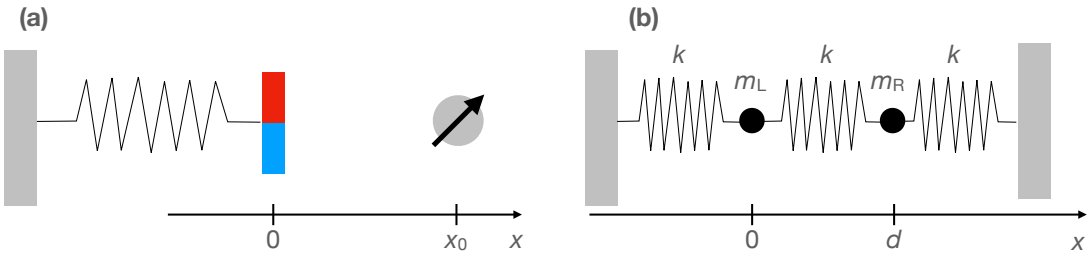


Figure 2: Coupled quantum systems. (a) A harmonic oscillator coupled to a spin. (b) Two harmonic oscillators coupled to each other.

2. Hamiltonian:

$$H = \frac{1}{2} g\mu_B B_{\text{mag}}(x_0)\sigma_z. \tag{19}$$

3. Assume now that the magnet is attached to a spring, such that it can move along x. Denote the displacement of the magnet with y. In equilibrium, $y = 0$.

4. Composite Hamiltonian of moving magnet and non-moving spin:

$$H = \hbar\omega_{\text{osc}} \left(a^\dagger a + 1/2\right) + \frac{1}{2} g\mu_B B_{\text{mag}}(x_0 - y)\sigma_z \tag{20}$$

Note that $y = \frac{\ell}{\sqrt{2}}(a + a^\dagger)$, where $\ell = \sqrt{\frac{\hbar}{m\omega_{\text{osc}}}}$, and m is the mass of the magnet.

5. Note that the tensor product structure of the operators in Eq. (20) is suppressed, 'implicit'. Could you write out Eq. (20) such that the tensor-product structure is explicit?

6. Assuming that $y \ll x_0$, we can Taylor-expand B_{mag}:

$$H = \hbar\omega_{\text{osc}} \left(a^\dagger a + 1/2\right) + \frac{1}{2} g\mu_B B_{\text{mag}}(x_0)\sigma_z - y\frac{1}{2} g\mu_B \left.\frac{dB_{\text{mag}}}{dx}\right|_{x=x_0} \sigma_z \tag{21}$$

7. Hence, the Hamiltonian is written as

$$H = H_{\text{osc}} + H_{\text{spin}} + H_{\text{int}}, \tag{22}$$

with

$$H_{\text{int}} = -\frac{1}{2} g\mu_B B'_{\text{mag}} \frac{\ell}{\sqrt{2}}(a + a^\dagger) \otimes \sigma_z. \tag{23}$$

4.2 Two coupled harmonic oscillators

1. Figure 2(b) depicts two masses that can move on a line, i.e., along the x axis. Without the central spring, the two masses would form two independent harmonic oscillators. The central spring makes these harmonic oscillators interact, i.e,. it 'couples' the oscillators.

2. We denote the coordinate of m_L as x_L, and the coordinate of m_R as x_R. In equilibrium, the springs are relaxed. Furthermore, in equilibrium, $x_L = 0$ and $x_R = d$. We denote the displacement of the masses from their equilibrium positions as $y_L = x_L$ and $y_R = x_R - d$.

3. Exercise: Based on the classical energy of this system, show that the Hamiltonian reads

$$H = \hbar\omega_L(a_L^\dagger a_L + 1/2) + \hbar\omega_R(a_R^\dagger a_R + 1/2) + H_{\text{int}}, \tag{24}$$

where $\omega_L = \sqrt{\frac{2k}{m_L}}$, $\omega_R = \sqrt{\frac{2k}{m_R}}$, and

$$H_{\text{int}} = -ky_Ly_R = -\frac{k\ell_L\ell_R}{2}\left(a_L + a_L^\dagger\right) \otimes \left(a_R + a_R^\dagger\right). \tag{25}$$

Furthermore, $\ell_L = \sqrt{\frac{\hbar}{m_L\omega_L}}$ and $\ell_R = \sqrt{\frac{\hbar}{m_R\omega_R}}$.

4. Remark: often the interaction Hamiltonian is approximated as

$$H_{\text{int}} \approx -\frac{k\ell_L\ell_R}{2}\left(a_L \otimes a_R^\dagger + a_L^\dagger \otimes a_R\right). \tag{26}$$

This approximation is also called 'rotating wave approximation' (RWA). We state without proof that it is a reasonable approximation if $k \ll m_L\omega_L^2, m_R\omega_R^2$ (weak coupling) and near resonance $\omega_L \approx \omega_R$.